Fiscal Year:	FY 2014	**Task Last Updated:**	FY 10/16/2013
PI Name:	Strangman, Gary E Ph.D.		
Project Title:	Multi-Use Near-Infrared Spectroscopy System for Spaceflight Health Applications		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Smart Medical Systems and Technology Team		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **ExMC**:Exploration Medical Capabilities		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	strang@nmr.mgh.harvard.edu	**Fax:**	FY
PI Organization Type:	NON-PROFIT	**Phone:**	617-724-0662
Organization Name:	Massachusetts General Hospital		
PI Address 1:	Department of Psychiatry		
PI Address 2:	149 13th Street, Suite 2651		
PI Web Page:			
City:	Charlestown	**State:**	MA
Zip Code:	02129-2020	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Crew Health NNJ11ZSA002NA
Start Date:	10/01/2012	**End Date:**	09/30/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	DiPasquale, Dana (Massachusetts General Hospital) Zhang, Quan (Massachusetts General Hospital)		
Grant/Contract No.:	NCC 9-58-SMST02801		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	While ultrasound has proved invaluable for a wide variety of medical assessments during spaceflight, it is largely blind to two important aspects of human physiology: blood oxygenation and blood volume. Near-infrared spectroscopy (NIRS), in contrast, is highly sensitive to blood oxygenation and volume, and hence has many medical uses, ranging from the gold standard of pulse oximetry, to tissue oxygenation and perfusion measurements, to imaging of brain function, hemorrhage or ischemia. NIRS could thus provide an excellent complement to ultrasound for spaceflight medical use. Importantly, a single NIRS system could support all of the above applications, while using non-ionizing near-infrared light, and remaining non-invasive, low power, and highly portable. Currently, the only NIRS-type measurement available in flight is pulse oximetry. Existing NIRS-based instruments are either ill-suited to spaceflight (large, heavy, and consume too much power), or are incapable of imaging. Our central objective is to develop and test a new NIRS system, NINscan M: a multi-use, wearable, and battery powered device with three distinct NIRS capabilities: (1) imaging of regional tissue oxygenation, perfusion and hemodynamics, including brain function, (2) point measurements of oxygenation suitable for muscle assessment, and (3) tissue oximetry. These capabilities can fill various key roles in spaceflight medical assessment, and could help address gaps in at least seven Human Research Roadmap risk categories: recognizing injured or ill crewmembers, muscle status monitoring, orthostatic intolerance assessment, detection or monitoring of behavioral or psychiatric conditions, effects of radiation on the central nervous system, sleep loss, and cardiac effects of spaceflight. The NINscan M design will be an embedded microcontroller system based upon our laboratory's series of prototype NIRS devices, which have been designed for mobile and long-duration monitoring. In addition to the three NIRS capabilities above, the system will also include the ability to record synchronized, auxiliary data streams for monitoring cardiac and skeletal muscle electrical activity, motion, force production, and temperature to further enhance its diagnostic value. The overall system design will be modular, such that the NIRS and auxiliary sensors can be flown and used singly or in combination, as indicated by medical operational requirements. To minimize the data management burdens for astronauts, NINscan M will be designed to seamlessly integrate with the NSBRI-supported SmartMED platform for automatically detecting, consolidating, storing, communicating and displaying biomedical and environmental data. Once developed, we will thoroughly characterize the NINscan M system, and perform human testing for: (1) sensitivity and specificity to changes in blood volume and oxygenation when imaging skin perfusion (shallow layers) and imaging brain function (deep tissue), (2) the ability to quantify muscle force production and oxygen extraction to assess muscle performance, and (3) validity in making tissue oximetry measurements. During development, we will work with ISSMP to ensure the resulting system is designed and developed within the constraints and requirements for spaceflight hardware and software. NINscan M will thus provide new capabilities, complementary to those currently available in-flight, which are important for medical assessments across a wide variety of spaceflight medical conditions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Currently, pulse oximetry is available onboard the ISS, and one would expect it to be flown on future exploration class missions. A pulse oximeter, however, provides only a small subset of the information available from a full NIRS measurement. In particular, whereas pulse oximetry can only provide oxygenation measures in the arterial compartment when a sufficiently strong pulse is available, NIRS measurements can provide arterial measurements as well as venous and whole-tissue measurements, and does not require clear pulse signals (e.g., compartment syndrome, during muscle contractions, or in patients with a weak or thready pulse). A NIRS-based imager can further provide spatial information about tissue oxygenation or perfusion. This can help determine, for example: (1) the relative conditioning status of different but adjacent muscle groups, or (2) assessing tissue oxygenation in different tissue layers (e.g., scalp versus cerebral oxygenation, or skin versus fat versus deeper muscle layers), or (3) identifying the location or size of an internal hemorrhage. Moreover, when deploying a sufficient number of near-infrared wavelengths and source/detector locations, one can further measure water content (e.g., edema), and correct measurements of deeper tissue layers for skin color, fat layers, and dynamic changes in systemic physiology (cardiac, respiratory, and other vasomotor interference). These capabilities of NIRS over standard pulse oximetry are not currently available in spaceflight but would be provided by NINscan M, as proposed in this project. The oxygenation and perfusion measures just described can also be enhanced by various auxiliary measurements. For example, electrocardiography (ECG) can be combined with systemic NIRS measurements to help estimate cardiac output. Electromyography (EMG) can be combined with muscle oxygenation measures to better understand muscle endurance and conditioning. Accelerometry or temperature measures can help identify and even compensate for environmental influences on NIRS measurements. Our goal with NINscan M is to provide not only shallow- and deep-tissue NIRS imaging capabilities, but also the ability to record various auxiliary measures—each useful in their own right—to increase NIRS assessment capabilities. All NIRS and auxiliary functions will be provided modularly, so that missions with different requirements need only pack and deploy the necessary components. Importantly, the system will integrate with the SmartMED platform to minimize risks associated with astronaut time and training burdens arising from data collection and management.
Task Progress:	We proposed to develop the first wearable NINscan-M imaging system during the first 16 months of this project. As of the completion of month 11, we have a functioning NINscan-M recorder, which includes two double-sided circuit boards plus one imaging sensor. The device provides 64 channels of data from eight laser diodes and eight monolithic photodiodes. The device design and capabilities are as follows: Light Sources/Detectors: The eight lasers (four dual-color diodes at 780 and 830nm) are turned on using a time-division multiplexing scheme to minimize power use. Each photodiode has its own variable gain setting for wide dynamic range. The near-infarared spectroscopy (NIRS) data is collected with 16-bit resolution at a 250Hz sampling rate. A preliminary test shows current consumption is approximately 200mA, which would provide a battery life of more than 20 hours. Imaging Sensor: A testing imaging sensor with a size of 70mm x 30mm has been built with 4 source locations and 7 detector locations. The source-to-detector separations are 15mm, 20mm, 25mm, or 40mm. On a physiologically realistic optical phantom, we have measured a 26:1 signal to noise ratio at the 40mm separation. Auxiliary Inputs: An 8-channel analog input for bio-potential signals (e.g., EMG, ECG, and/or EEG) is also included, which can be collected with 24-bit resolution at a 250Hz sampling rate. A digital accelerometer, temperature module, and a force sensor used to sense force up to 100lb have been tested and are ready to be integrated into the system. User Interface: The imaging device has an OLED screen and four buttons to provide a user interface. A digital output is included to synchronize external devices when needed. The raw data are stored on an SD card convenient for transferring and postprocessing. A Bluetooth module on the board can also be activated for wireless data streaming.

	Integration with the SmartMED Communication Platform: The Bluetooth module has been added to the imager, which will allow digital control of machine parameters and scanning mode as well as wireless data streaming for integration with the SmartMED platform. The current system represents the smallest, lightest, and lowest-power NIRS imaging system currently in existence, with further size reductions expected. While full characterization, human testing, and validation will be required, the flexible design will enable deployment in a variety of settings or for a variety monitoring needs ranging from general medical risks, to cardiovascular risks, to musculoskeletal risks, to radiation risks, to behavioral and performance risks. In addition to device development, Drs. Strangman and Zhang published two papers quantifying the depth-sensitivity of NIRS in neuroimaging studies and the role of overlying tissue layers (scalp and skull) on NIRS measurements. In addition, a suite of computational tools are being developed to handle the data to be generated by NINscan-M.
Bibliography Type:	Description: (Last Updated: 07/08/2026)
Abstracts for Journals and Proceedings	Hu G, Zhang Q, Strangman GE. "NINscan M: Multi-use near-infrared imaging for spaceflight health monitoring." 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. , Feb-2013
Articles in Peer-reviewed Journals	Strangman GE, Li Z, Zhang Q. "Depth sensitivity and source-detector separations for near infrared spectroscopy based on the Colin27 brain template." PLoS One. 2013 Aug 1;8(8):e66319. http://dx.doi.org/10.1371/journal.pone.0066319 ; PubMed PMID: 23936292 , Aug-2013
Articles in Peer-reviewed Journals	Strangman GE, Zhang Q, Li Z. "Scalp and skull influence on near infrared photon propagation in the Colin27 brain template." Neuroimage. 2013 May 7. In Press, Corrected Proof, Available online 7 May 2013. http://dx.doi.org/10.1016/j.neuroimage.2013.04.090 , May-2013